Practice Set 10

① Find the Laplace transform of

$$x(t) = e^{-t^2} \quad \text{for all } t.$$

You can use the fact :

$$\int_{-\infty}^{+\infty} e^{-(t-a)^2} dt = \sqrt{\pi} \quad \text{for any } a \in \mathbb{C}.$$

② In question ① we saw a two-sided signal e^{-t^2} whose ROC is the entire s-plane.

Now consider another two-sided signal

$$x(t) = \frac{1}{1+t^2} \quad \text{for all } t.$$

(a) Find the ROC of this signal.

(b) Find the ROC of :

$$e^{-t^2} u(t) + \frac{1}{1+t^2} u(-t)$$

Is the boundary line a part of the ROC or not ?

(3) (Examples 9-9, 9.10, 9.11).

For $X(s) = \dfrac{1}{(s+1)(s+2)}$,

(a) Find all possible ROCs.

(b) For each of these ROCs find the inverse Laplace transform, i.e. find $x(t)$.

(4) Find the Laplace transform of:

(a) $\sin(\omega_0 t + \phi_0)\, u(t)$

(b) $e^{-a|t|}$ where a is real and $a > 0$.

(c) $e^{-a|t-t_0|}$, where $a > 0$.

(5) (Example 9.20 from OWN).

(a) If $h(t)$ is the impulse response of an LTI system, and this system is stable, then:

does the $j\omega$ axis of the s-plane belong to the ROC ?

(b) If $H(s) = \dfrac{s-1}{(s+1)(s-2)}$ and if the system is stable, identify the ROC.

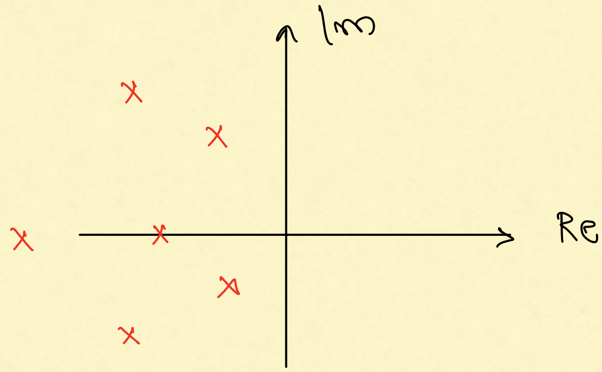
(c) Find $h(t)$ assuming that the system is stable.

(d) Again, if $H(s) = \dfrac{(s-1)}{(s+1)(s-2)}$, and if the system is stable, what can we say about the causality of this system?

(e) What if $H(s)$ is as before, but now we are told that this LTI system is causal, what is $h(t)$ in this case?

6 Suppose $h(t)$ is the impulse response of an LTI system that is both stable and causal. Also, suppose $H(s)$ is rational.

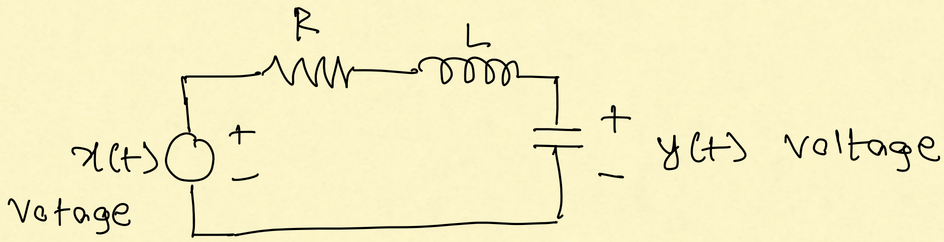
Argue that every pole of $H(s)$ has negative part.



Stable, Causal LTI system:

All poles have -ve real parts.

⑦ Based on Example 9.24 of OWN (& Lec 27 of EE1101)



The differential equation describing this system is:

$$\frac{d^2y}{dt^2} + \frac{R}{L}\frac{dy}{dt} + \frac{1}{LC}y(t) = \frac{1}{LC}x(t).$$

(a) Apply Laplace transform to either side of this equation and find $\frac{Y(s)}{X(s)}$.

Remark: Since this is an LTI system:

$$y = x * h \Rightarrow Y(s) = X(s) \cdot H(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)}.$$

(b) Denote $\sqrt{\frac{1}{LC}} = \omega_0$ and $\varepsilon = \frac{R}{2}\sqrt{\frac{C}{L}}$

Write $H(s)$ in terms of ε and ω_0

(c) Find all the poles of $H(s)$ in terms of ε and ω_0. Note that $\varepsilon, \omega_0 > 0$.

(d) Consider the following cases :

(i) $\varepsilon > 1$ (called overdamped case)

Show that both poles are: real and negative, and are distinct.

(ii) $\varepsilon = 1$ (called critically damped case)

Show that both poles are: identical, real and negative.

(iii) $\varepsilon < 1$ (called underdamped case)

Show that both poles are: non-real, have negative real part and are conjugates of each other.

(use $\omega_d = \omega_0\sqrt{1-\varepsilon^2}$ in your calculations).

(e) This circuit is a causal LTI system, and hence, $h(t)$ is a right-sided signal.

Find $h(t)$ for each of the three cases:

$\varepsilon > 1$, $\varepsilon = 1$ and $\varepsilon < 1$.

(f) What is $h(t)$ if $\xi = 0$ and $\omega_0 > 0$?

This is the undamped system.

(8) Exercise problems from OWN :

9.6, 9.9,

9.22 (a), (b), (c), (g)